



**Data
Sources**



**Data
ingestion**



**Data
transmission**



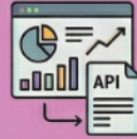
**Data
preprocessing**



**Data
loading**



**Data
storage**



**Data
sinks**

Technical and interface heterogeneity

Connectivity instability

Semantic and time-management fragility

Operational and staffing limitations

Economic and change pressure